Iterated Symmetric Centipede: Learning and Evolutionary Games

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March 10, 2026

Abstract

We study the *Iterated Symmetric Centipede* (ISC) game. The ISC involves repeated plays of the *One-shot Symmetric Centipede* (OSC), which is the "classic" Centipede game with random selection of the first moving player. Since ISC involves repeated plays of OSC, we can use *history-dependent* strategies, similar to the All-C, All-D, Grim etc. used in the *Iterated Prisoner's Dilemma*. We are especially interested in the interaction between these strategies. We first study the Nash equilibria of the ISC using standard game theoretic methods. Then we consider an *evolutionary game* using the ISC as the "base game". We study the evolutionary game for (i) the case of large populations, using *replicator dynamics* and (ii) for the case of *small populations* using Markov chains (obtained from the use of *pairwise proportional imitation*). We see that the presence of cooperation-promoting strategies (such as Grim) favors cooperation. A *learning theory* interpretation of replicator dynamics and pairwise proprtional imitation is also presented.

1 Introduction

In this paper we study games derived from the *Centipede game*, an extensive form game of perfect information. The "classic" Centipede game was introduced in [15]; a recent overview of *experimental* approaches to the field appears in [10]. In the current paper we study normal form variants, especially the *iterated symmetric centipede* (ISC), which is obtained by the following procedure.

First, using a standard construction, we obtain a normal form game which is equivalent to the "classical" (tree) Centipede. To remove the asymmetry between first and second player of the original Centipede we modify the game by randomizing the selection of the player who has the first move; in this way we obtain the one-round *symmetric* centipede (OSC). This has been previously studied by several authors; e.g., see [3, p.238, pp.255-260] and [4, 5] for a general analysis, [17, 8, 14] for an evolutionary games approach and [11] for the use of genetic algorithms.

However, as mentioned, we are mainly interested in the iterated symmetric centipede (ISC) game, which consists of playing T rounds of the OSC. We assume that, in every round of ISC, both players recall the moves and outcome of the previous rounds. This introduces the possibility of using *history-dependent* strategies; our main interest in the current paper is the effect of using such strategies to promote cooperation and punish defection.

Hence we adopt an interpretation of repeated Centipede play at the *round-level*, where each round consists of a complete one-shot Centipede "tournament" and strategies update only *between rounds*. Under this interpretation it is natural to lift strategies from the *Iterated Prisoner's Dilemma* (IPD) context, such as Tit-for-Tat and Grim Trigger. To produce the current-round behavior, such strategies use the opponent's actions in previous rounds (rather than simply reacting to within-round decisions). In short, the subject of our paper can be understood as an *evolutionary tournament* of the type previously used to study the IPD (most famously by Axelrod in [1]).

As far as we can tell, this approach has not been previously reported in the literature. The closest we have found is several works which apply various learning and / or evolutionary frameworks to the Centipede [8, 12, 13, 17, 18, 19, 20]. For example, in [20] Centipede play is modeled as a stochastic learning process in which players update strategies between rounds based on opponents' prior behavior; these learning dynamics capture adaptive round-level responses consistent with Tit-for-Tat–type reciprocal strategies, even when players do not follow deterministic trigger rules. The other cited papers follow similar approaches, usually applied to the extensive form (and often *un*symmetric) Centipede.

The current paper is structured as follows. In Section 2 we briefly review the "classic" (tree) Centipede and the normal form one-shot symmetric Centipede. In Section 3 we present the iterated symmetric centipede game (ISC). In Section 4 we identify the Nash equilibria (NE) of ISC, in Section 5 we study the replicator dynamics of an evolutionary game with ISC as the base game and in Section 6 we study the evolutionary stable strategies. In Section 7 we present a Markovian analysis of the *small population* evolutionary game. In Section 8 we discuss the connections to learning theory. Finally, in Section 9 we summarize our results and present future research directions.

2 Preliminaries

2.1 "Classic" Centipede

The game of centipede has been introduced in [15]. In subsequent works, many variants of the original game have been introduced, but they all share some basic characteristics.

1. It is an extensive form game of perect information (a tree game).

2. The game tree has the "centipede shape", namely it consists of a path of M vertices, with each path vertex also having an additional edge ending at a leaf vertex. Hence, the game tree contains a total of $2M$ vertices; we denote the path vertices as $1, 2, ..., M$ and the leaves as $M+1, M+2, ..., 2M$.

3. The terminal vertices are $M+1$, $M+2$, ..., $2M$; the $(M+m)$-th vertex has payoffs (a_m, b_m).

4. The payoffs satisfy:

$$a_1 > a_2, a_3 > a_4, a_5 > a_6, ...$$
$$b_1, b_2 > b_3, b_4 > b_5, b_6 > b_7, ...$$

The game is played as follows:

1. P_1 plays at odd turns, P_2 plays at even turns.
2. The sum of payoffs at the t-th turn is $c_t = a_t + b_t = f(t)$, where f is an increasing function of t.
3. The c_t is divided as follows: if the moving player goes to the leaf vertex, he receives pc_t and the other player receives $(1-p)\,c_t$, where $p \in (\frac{1}{2}, 1]$.

In this paper we will always take $f(t) = t$. As an example, taking $M = 4$ and $p = \frac{3}{4}$, we will have

t	1	2	3	4
c_t	1.00	2.00	3.00	4.00
a_t	0.75	0.50	2.25	1.00
b_t	0.25	1.50	0.75	3.00

resulting in the game tree illustrated in Fig.1.

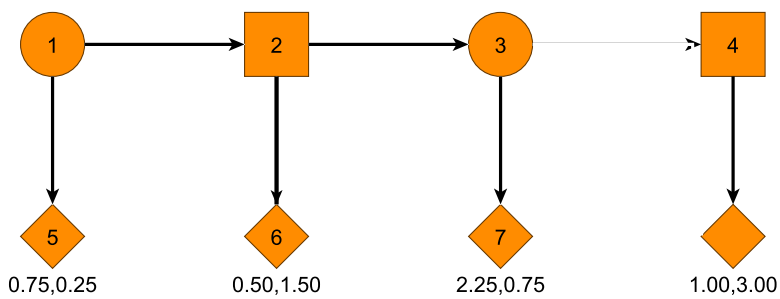

Figure 1: A Centipede Game

Hence at each nonterminal vertex, the player who has the move can decide to either terminate the game (move "down") or continue to the next stage (move "right"); this continues until a terminal (diamond) vertex is reached, at which point the game terminates unconditionally. In other words, whenever a player has the move he can choose to: (a) either get immediate payoff, (b) or continue the game hoping for a higher payoff at a later stage. Note that the *total* payoff increases with the length of the game and also that the terminating player receives a higher proportion of the total payoff.

Solving the game by *backward induction* we obtain the unique subgame perfect equilibrium. This, in the example of Fig. 1, is for the first player to immediately move down, resulting to payoff vector $(0.75, 0.25)$. However this is counterintuitive: if both players keep moving right, they will end up with payoffs $(1.00, 3.00)$. I.e., they would both benefit by using the strategy of always moving to the right; this however is not an equilibrium.

Let us now introduce a family of *strategies* $\rho_1, ..., \rho_M$ which can be used by both players. These strategies are defined as follows:

$\forall m : \rho_m =$ "I terminate the game as soon as it reaches or passes the m-th stage and it is my turn to play."

Let us denote the two players by P_1 and P_2 and define for $i \in \{1,2\}$ and $m,n \in \{1,...,M\}$ the following:

$$Q_i(\rho_m, \rho_n) = \text{“Payoff to } P_i \text{ when } P_1 \text{ uses } \rho_m \text{ and } P_2 \text{ uses } \rho_n\text{”}.$$

Now let us compute some examples $Q_1(\rho_m, \rho_n)$, $Q_2(\rho_m, \rho_n)$, for the example game of Fig. 1.

1. Suppose that both players use ρ_1. Hence P_1 terminates the game at $t = 1$; the total payoff is $Q(\sigma_1, \sigma_2) = 1$ and it is split between the two players as follows: $Q_1(\rho_1, \rho_1) = \frac{3}{4} \cdot 1 = 0.75$ and $Q_2(\rho_1, \rho_1) = \frac{1}{4} \cdot 1 = 0.25$.

2. Suppose that P_1 uses ρ_2 and P_2 uses ρ_4. At $t = 2$, P_1's threshold is reached but he does not have the move, while P_2's threshold has not been yet reached; at $t = 3$, P_1's threshold is exceeded and he has the move (while P_2's threshold has not been reached yet) hence P_1 terminates the game. The total payoff is $Q(\rho_2, \rho_4) = 3$ and it is split between the two players as follows: $Q_1(\rho_2, \rho_4) = \frac{3}{4} \cdot 3 = 2.25$ and $Q_2(\rho_2, \rho_4) = \frac{1}{4} \cdot 3 = 0.75$.

3. Suppose that P_1 uses ρ_4 and P_2 uses ρ_3. At $t = 3$, P_1's threshold has not been reached, while P_2's threshold has been reached but he does not have the move; at $t = 4$, P_1's threshold has been reached and P_2's threshold is exceeded, hence P_2 (having the move) terminates the game. The total payoff is $Q(\rho_2, \rho_4) = 4$ and it is split between the two players as follows: $Q_1(\rho_2, \rho_4) = \frac{1}{4} \cdot 4 = 1.00$ and $Q_2(\rho_2, \rho_4) = \frac{3}{4} \cdot 4 = 3.00$.

2.2 One-Round Symmetric Centipede

We can employ a standard procedure to obtain, from the tree Centipede, a normal form bimatrix game, which we will call the *One-Round Symmetric Centipede* (OSC) game. We start by keeping the assumptions of the tree Centipede (total payoff at t is t, P_1 receives pt, P_2 receives $(1-p)t$). Further, we define $M \times M$ matrices $A^{(1)}$ and $A^{(2)}$ as follows:

$$\forall i \in \{1,2\} : A^{(i)}_{mn} = \text{“Payoff to } P_1 \text{ when: } P_1 \text{ plays } \rho_m,\ P_2 \text{ plays } \rho_n \text{ and } P_i \text{ plays first”}.$$

Now we suppose that P_1 plays first and we compute $A^{(1)}$. We reason as follows.

1. When i is odd and j is even:

$$A^{(1)}_{i,j} = \begin{cases} p \min(i,j), & \text{if } i < j, \\ (1-p)\min(i,j), & \text{if } i > j. \end{cases}$$

2. When both i and j are odd, $A^{(1)}_{i,j} = A^{(1)}_{i,j+1}$, since the second player will not play until we are in an even turn and the turn is equal or above j.

3. Similarly, when both i and j are even, $A^{(1)}_{i,j} = A^{(1)}_{i+1,j}$.

4. Finally, when i is even and j is odd, $A^{(1)}_{i,j} = A^{(1)}_{i+1,j+1}$.

Similar properties are also true for $A^{(2)}_{i,j}$. Letting A be the expected payoff matrix of OSC, we have

$$A = \frac{1}{2}\left(A^{(1)} + A^{(2)}\right).$$

Dropping the $\frac{1}{2}$ factor (which will not influence our later results), we more specifically have:

$$\begin{aligned}
\text{for } i \text{ odd, } j \text{ even: } A_{i,j} &= A^{(1)}_{i,j} + A^{(2)}_{i+1,j+1},\\
\text{for } i \text{ even, } j \text{ odd: } A_{i,j} &= A^{(1)}_{i+1,j+1} + A^{(2)}_{i,j},\\
\text{for } i \text{ odd, } j \text{ odd: } A_{i,j} &= A^{(1)}_{i,j+1} + A^{(2)}_{i+1,j},\\
\text{for } i \text{ even, } j \text{ even: } A_{i,j} &= A^{(1)}_{i+1,j} + A^{(2)}_{i,j+1}.
\end{aligned}$$

Let us compute some particular values.

1. For $A_{i,i}$ we have:

$$\begin{aligned}
\text{for } i \text{ odd: } A_{i,i} &= A^{(1)}_{i,i+1} + A^{(2)}_{i+1,i} = ip + i(1-p) = i,\\
\text{for } i \text{ even: } A_{i,i} &= A_{i,i} = A^{(1)}_{i+1,i} + A^{(2)}_{i,i+1} = i(1-p) + ip = i.
\end{aligned}$$

2. For $A_{i,1}$ with $i > 1$ we have:

$$\begin{aligned}
\text{for } i \text{ odd: } A_{i,1} &= A^{(1)}_{i+1,2} + A^{(2)}_{i,1} = 2(1-p) + (1-p) = 3(1-p),\\
\text{for } i \text{ even: } A_{i,1} &= A^{(1)}_{i,2} + A^{(2)}_{i+1,1} = 2(1-p) + s(1-p) = 3(1-p).
\end{aligned}$$

3. Similarly, we can calculate that:

$$A_{1,i} = 3p.$$

4. For $A_{i,j}$ with $i > j$:

$$\begin{aligned}
\text{for } i \text{ odd, } j \text{ even: } A_{i,j} &= A^{(1)}_{i,j} + A^{(2)}_{i+1,j+1} = j(1-p) + (j+1)(1-p) = (2j+1)\cdot(1-p),\\
\text{for } i \text{ even, } j \text{ odd: } A_{i,j} &= A^{(1)}_{i+1,j+1} + A^{(2)}_{i,j} = (j+1)(1-p) + j(1-p) = (2j+1)\cdot(1-p),\\
\text{for } i \text{ odd, } j \text{ odd: } A_{i,j} &= A^{(1)}_{i,j+1} + A^{(2)}_{i+1,j} = i(1-p) + (j+1)(1-p) = (2j+1)(1-p),\\
\text{for } i \text{ even, } j \text{ even: } A_{i,j} &= A^{(1)}_{i+1,j} + A^{(2)}_{i,j+1} = (j+1)(1-p) + j(1-p) = (2j+1)(1-p).
\end{aligned}$$

5. Similarly, we can calculate that for $i < j$:

$$A_{i,j} = (2i+1)p.$$

So finally we have

$$A = \begin{bmatrix}
1 & 3p & 3p & 3p & \dots & 3p\\
3\,(1-p) & 2 & 5p & 5p & \dots & 5p\\
3\,(1-p) & 5\,(1-p) & 3 & 7p & \dots & 7p\\
3\,(1-p) & 5\,(1-p) & 7\,(1-p) & 4 & \dots & 9p\\
\dots & \dots & \dots & \dots & \dots & \dots\\
3\,(1-p) & 5\,(1-p) & 7\,(1-p) & 9\,(1-p) & \dots & M
\end{bmatrix}.$$

3 Iterated Symmetric Centipede

Now we introduce the *Iterated Symmetric Centipede* (ISC) game. It is obtained from OSC in the standard manner which is used in the study of iterated games; i.e., ISC consists in playing OSC for T rounds. What makes ISC qualitatively different from OSC is that a player can use strategies with *memory*. In other words, the OSC strategy used in the t-th round of ISC may depend on the moves of previous rounds. Hence ISC has the same relationship to OSC as the Iterated Prisoner's Dilemma (IPD) has to Prisoner's Dilemma (PD). In this paper we concentrate on three ISC strategies:

1. $\sigma_{\text{All-}M}$**:** In every OSC game play ρ_M.
2. $\sigma_{\text{All-1}}$: In every OSC game play ρ_1.
3. σ_G or *Grim***:** Play ρ_M at the first round and, after that, as long as the opponent has played ρ_M in all previous rounds; if at some round the opponent plays ρ_m with $m < M$, then play ρ_1 in all subsequent rounds.

This strategy ensemble allows us to explore the interplay between the fully cooperating strategy $\sigma_{\text{All-}M}$, the fully defecting strategy $\sigma_{\text{All-1}}$ and the strategy σ_G which starts with cooperation but punishes defections. The analogy between $\sigma_{\text{All-}M}$, $\sigma_{\text{All-1}}$, σ_G and the IPD strategies All-C, All-D, Grim is obvious.

More precisely, we define an ISC game as follows: it is played between two players, each of which can play one of the pure strategies $\sigma_{\text{All-}M}$, $\sigma_{\text{All-1}}$, σ_G. At the beginning, each player chooses one of the pure strategies and uses it in playing a sequence of T rounds of OSC. Referring to $\sigma_{\text{All-}M}$, $\sigma_{\text{All-1}}$, σ_G as the first, second and third strategy, we compute the payoff matrix C as follows:

C_{ij} = "Payoff to a user of the i-th strategy whe playing against a user of the j-th strategy".

The C matrix is computed in terms of the A matrix of OSC game. Let us give some examples:

1. C_{11} is the payoff obtained by a $\sigma_{\text{All-}M}$ user playing against another $\sigma_{\text{All-}M}$ user; since he plays ρ_M against ρ_M for T rounds, his payoff equals TA_{MM} .
2. Similarly, C_{12} is the payoff obtained by a $\sigma_{\text{All-}M}$ user playing against a $\sigma_{\text{All-1}}$ user; since he plays ρ_M against ρ_1 for T rounds, his payoff equals to TA_{M1} .
3. Consider now C_{23}; this is the payoff of a σ_G user playing against a $\sigma_{\text{All-1}}$ user. In the first round, he plays ρ_M against ρ_1 and receives A_{M1}; in the remaining $T-1$ rounds he plays ρ_1 against ρ_1 and receives $(T-1)\,A_{11}$; so his total payoff is $A_{M1} + (T-1)\,A_{11}$.

The remaining entries of C are computed in similar manner, resulting in the following expression for C:

$$C = \begin{bmatrix} TA_{MM} & TA_{M1} & TA_{MM} \\ TA_{1M} & TA_{11} & A_{1M} + (T-1)\,A_{11} \\ TA_{MM} & A_{M1} + (T-1)\,A_{11} & TA_{MM} \end{bmatrix} = \begin{bmatrix} TM & 3\,(1-p)\,T & TM \\ 3pT & T & 3p + (T-1) \\ TM & 3\,(1-p) + (T-1) & TM \end{bmatrix}$$

Finally, the bimatrix ISC game is $D = \left(C, C^T\right)$.

4 Nash equilibria

Recall the definition of a *Nash Equilibrium* (NE).

Definition 4.1 A pair of strategies $(\widehat{\sigma}_1, \widehat{\sigma}_2)$ is called a *Nash equilibrium* of a two player game with payoffs Q_1, Q_2, iff

$$\forall \sigma_1 : Q_1(\widehat{\sigma}_1, \widehat{\sigma}_2) \geq Q_1(\sigma_1, \widehat{\sigma}_2) \text{ and } \forall \sigma_2 : Q_2(\widehat{\sigma}_1, \widehat{\sigma}_2) \geq Q_1(\widehat{\sigma}_1, \sigma_2).$$

Now we give a family of NE of ISC.

Proposition 4.2 For every $M \in \{3, 4, ...\}$ and large enough T, the following are NE of ISC for any $a', a'' \in [0, 1]$:

$$((1-a', 0, a'), (1-a'', 0, a'')).$$

Proof. Let $Q_i(\sigma', \sigma'')$ denote P_i's payoff under the strategy profile (σ', σ''). Consider the following cases. We are looking for NE of the form $((1-a, 0, a), (1-d, 0, d))$. We have

$$\begin{aligned}
&Q_1((1-a, 0, a), (1-d, 0, d)) \\
&= \begin{bmatrix} a & 0 & 1-a \end{bmatrix} \begin{bmatrix} TM & 3(1-p)T & TM \\ 3pT & T & 3p+(T-1) \\ TM & 3(1-p)+(T-1) & TM \end{bmatrix} \begin{bmatrix} d \\ 0 \\ 1-d \end{bmatrix} = TM
\end{aligned}$$

Suppose P_1 deviates to the strategy $(b, c, 1-b-c)$ with $b, c, b+c \in [0, 1]$. Then we have

$$\begin{aligned}
&Q_1((b, c, 1-b-c), (1-d, 0, d)) \\
&= \begin{bmatrix} b & c & 1-b-c \end{bmatrix} \begin{bmatrix} TM & 3(1-p)T & TM \\ 3pT & T & 3p+(T-1) \\ TM & 3(1-p)+(T-1) & TM \end{bmatrix} \begin{bmatrix} d \\ 0 \\ 1-d \end{bmatrix} \\
&= 3dcpT + 3cp - 3cpd + cT - cTd - c + cd + TM - TMc
\end{aligned}$$

and

$$\begin{aligned}
\delta Q_1 &= Q_1((1-a, 0, a), (1-d, 0, d)) - Q_1((b, c, 1-b-c), (1-d, 0, d)) \\
&= cT(-3pd - 1 + d + M) + (3cpd - 3cp + c - cd)
\end{aligned}$$

For large enough T, δQ_1 is nonnegative iff

$$-3pd - 1 + d + M \geq 0 \tag{4.1}$$

and (4.1) is equivalent to one of the following:

$$\text{when } d > 0 : p \leq \frac{M+d-1}{d} = 1 + \frac{M-1}{d} \tag{4.2}$$

$$\text{when } d = 0 : M - 1 \geq 0 \tag{4.3}$$

Since both (4.2) and (4.3) are always true, δQ_1 is nonnegative for all admissible values of b and c; by an identical argument, the same is true for the corresponding δQ_2. Hence all $((1-a, 0, a), (1-d, 0, d))$ are NE. ■

The following proposition characterizes the *symmetric* NE of ISC.

Proposition 4.3 For every $M \in \{3, 4, ...\}$ and large enough T, two symmetric NE of ISC are

$$((1,0,0),(1,0,0)) \text{ and } ((0,0,1),(0,0,1)). \tag{4.4}$$

When $p \geq \frac{2}{3}$ the following are also symmetric NE of ISC:

$$((0,1,0),(0,1,0)), \tag{4.5}$$

$$\left(\left(\frac{3p-2}{M-2}, \frac{M-3p}{M-2}, 0\right), \left(\frac{3p-2}{M-2}, \frac{M-3p}{M-2}, 0\right)\right), \tag{4.6}$$

$$\left(\left(0, \frac{T(M-1)+1-3p}{T(M-1)-1}, \frac{3p-2}{T(M-1)-1}\right), \left(0, \frac{T(M-1)+1-3p}{T(M-1)-1}, \frac{3p-2}{T(M-1)-1}\right)\right). \tag{4.7}$$

These are the only symmetric NE of ISC.

Proof. The NE of (4.4) are special cases of the NE of Proposition 4.2. Now let $Q_i(\sigma', \sigma'')$ denote P_i's payoff under the strategy profile (σ', σ'') and consider the following cases.

For (4.5) we have $Q_1((0,1,0),(0,1,0)) = T$. Suppose P_1 switches to strategy $\tau = (a, 1-a-b, b)$. Then

$$\begin{aligned} Q_1(\tau,(0,1,0)) &= \begin{bmatrix} a & 1-a-b & b \end{bmatrix} \begin{bmatrix} TM & 3(1-p)T & TM \\ 3pT & T & 3p+(T-1) \\ TM & 3(1-p)+(T-1) & TM \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \\ &= 2aT - 3aTp + T + 2b - 3bp \end{aligned}$$

and

$$\begin{aligned} \delta Q &= Q_1((0,1,0),(0,1,0)) - Q_1(\tau,(0,1,0)) \\ &= -2aT + 3aTp - 2b + 3bp = aT(-2+3p) - 2b + 3bp \end{aligned}$$

For large enough T, the sign of δQ is determined by $aT(-2+3p)$, which (since $a > 0$) is nonnegative for $p \geq \frac{2}{3}$. Hence $((0,1,0),(0,1,0))$ is a NE iff $p \geq \frac{2}{3}$.

Next, let us look for NE of the form $((a, 1-a, 0),(c, 1-c, 0))$. In this case we get

$$\begin{aligned} Q_1(a,c) &= TMca + 3Tcp + 2Ta - 2Tac - 3Tap + T - Tc \\ Q_2(a,c) &= TMca + 3Tap + 2Tc - 2Tac - 3Tcp + T - Ta \end{aligned}$$

and solving

$$\begin{aligned} \frac{dQ_1}{da} &= TMc + 2T - 2Tc - 3Tp \\ \frac{dQ_2}{dc} &= TMa + 2T - 2Ta - 3Tp \end{aligned}$$

we get

$$a = c = \frac{3p-2}{M-2}.$$

These will be admissible iff $p \geq \frac{2}{3}$, in which case we get the NE

$$\left(\left(\frac{3p-2}{M-2}, \frac{M-3p}{M-2}, 0\right), \left(\frac{3p-2}{M-2}, \frac{M-3p}{M-2}, 0\right)\right).$$

Next, let us look for NE of the form $((0, a, 1-a), (0, c, 1-c))$. In this case we get

$$Q_1(a, c) = 2c - 3cp + 3ap + cT - a - ca + aT - aTc + TM - TMc - TMa + cTMa$$
$$Q_2(a, c) = 2a - 3ap + 3cp + cT - c - ca + aT - aTc + TM - TMc - TMa + cTMa$$

and solving

$$\frac{dQ_1}{da} = 3p - 1 - c + T - Tc - TM + cTM = 0$$
$$\frac{dQ_2}{dc} = 3p + T - 1 - a - aT - TM + TMa = 0$$

we get

$$a = c = \frac{TM - T - 3p + 1}{TM - T - 1}.$$

We always have $TM - T - 1 > 0$. Furthermore,

$$TM - T - 3p + 1 \geq TM - T - 2 = T(M-1) - 2$$

which, for large T, is nonnegative. Hence $a = c \geq 0$. Furthermore:

$$\forall p \in [0, 1] : \frac{TM - T - 3p + 1}{TM - T - 1} \in \left[\frac{TM - T - 2}{TM - T - 1}, \frac{TM - T + 1}{TM - T - 1}\right] \subseteq [0, 1].$$

Hence, since $p \geq \frac{2}{3}$,

$$\left(\left(0, \frac{T(M-1) - 3p + 1}{T(M-1) - 1}, \frac{3p-2}{T(M-1) - 1}\right), \left(0, \frac{T(M-1) - 3p + 1}{T(M-1) - 1}, \frac{3p-2}{T(M-1) - 1}\right)\right)$$

is a NE of ISC.

Finally, let us look for interior NE of the form $((a, b, 1-a-b), (c, d, 1-c-d))$. Computing $Q_1(a, b, c, d)$ and solving

$$\frac{dQ_1}{da} = -3dTp + 3dp - 2d + 2dT = 0$$
$$\frac{dQ_2}{db} = 3cpT + TMd - 1 + T + 3p - d + c - TM - 3pc - Tc - dT = 0$$

we get

$$c = \frac{1 - 3p - T + TM}{1 - 3p - T + 3Tp}, \quad d = 0.$$

For large T we have

$$c \simeq \frac{M}{3p} \geq 1.$$

Hence we do no get any NE (except for the pure NE $((1, 0, 0), (1, 0, 0))$ when $M = 3$ and $p = 1$). This completes the proof. ■

Example 4.4 Take $M = 4$, $p = \frac{3}{4}$, $T = 10$. Then we have

$$C = \begin{bmatrix} 40 & \frac{30}{4} & 40 \\ \frac{90}{4} & 10 & \frac{45}{4} \\ 40 & \frac{39}{4} & 40 \end{bmatrix}$$

and we get the following NE:

$$((0,1,0),(0,1,0)), \quad ((1-a',0,a'),(1-a'',0,a'')), \quad \left(\frac{1}{8},\frac{7}{8},0\right), \quad \left(0,\frac{115}{116},\frac{1}{116}\right).$$

However, with $M = 4$, $p = \frac{3}{5}$ and $T = 10$ we have

$$C = \begin{bmatrix} 40 & 12 & 40 \\ 18 & 10 & \frac{54}{5} \\ 40 & \frac{51}{5} & 40 \end{bmatrix}$$

and we only get the NE:

$$((1-a',0,a'),(1-a'',0,a''))$$

5 Replicator Dynamics

Consider an *evolutionary game* [6, 16, 17] involving a large population of players, playing ISC as the base game and using the strategies $\sigma_{\text{All-}M}$, $\sigma_{\text{All-1}}$, σ_G; let x_1 be the proportion of players using $\sigma_{\text{All-}M}$, x_2 be the proportion of players using $\sigma_{\text{All-1}}$ and x_3 be the proportion of players using σ_G. Furthermore, suppose that the x_m's evolve according to the *replicator dynamics* [6]:

$$\forall m : \frac{dx_m}{dt} = x_m \left((Cx)_m - x^T C x\right). \tag{5.1}$$

In our case (5.1) becomes

$$\frac{dx_1}{dt} = x_1 \left(TMx_1 + 3Tx_2 - 3Tx_2p + TMx_3 - \left(TMx_1^2 + 2x_1TMx_3 + 3x_2x_1T + Tx_2^2 + x_2x_3 + 2x_2Tx_3 + TM\right.\right.$$
$$\frac{dx_2}{dt} = x_2 \left(3pTx_1 + Tx_2 + 3x_3p + Tx_3 - x_3 - \left(TMx_1^2 + 2x_1TMx_3 + 3x_2x_1T + Tx_2^2 + x_2x_3 + 2x_2Tx_3 + TMx\right.\right.$$
$$\frac{dx_3}{dt} = x_3 \left(TMx_1 + 2x_2 - 3x_2p + Tx_2 + TMx_3 - \left(TMx_1^2 + 2x_1TMx_3 + 3x_2x_1T + Tx_2^2 + x_2x_3 + 2x_2Tx_3 + T\right.\right.$$

Proposition 5.1 For every $M \in \{3, 4, ...\}$ and every T, the following are admissible rest points of (5.1):

$$\forall x_1 \in [0,1] : (x_1, 0, 1-x_1), \tag{5.2}$$

$$((0,1,0),(0,1,0)). \tag{5.3}$$

When $p \geq \frac{2}{3}$, the following are also admissible rest points of (5.1):

$$\left(\frac{3p-2}{M-2}, \frac{M-3p}{M-2}, 0\right), \tag{5.4}$$

$$\left(0, \frac{T(M-1)+1-3p}{T(M-1)-1}, \frac{3p-2}{T(M-1)-1}\right). \tag{5.5}$$

These are the only rest points of (5.1) in the simplex

$$S_3 = \{(x_1, x_2, x_3) : x_1 + x_2 + x_3 = 1,\ x_i \in [0,1]\}.$$

Proof. To find the rest points we solve the system

$$\forall m : 0 = x_m \left((Cx)_m - x^T C x\right).$$

The solutions are:

$$\begin{aligned}
\text{always admissible: } & (0,1,0), \quad (x_1, 0, 1-x_1), \\
\text{admissible when } p \geq \frac{2}{3}\text{: } & \left(\frac{3p-2}{M-2}, \frac{M-3p}{M-2}, 0\right), \quad \left(0, \frac{TM-T-3p+1}{TM-T-1}, \frac{3p-2}{TM-T-1}\right), \\
\text{inadmissible: } & (-x_1, 0, 1-x_1).
\end{aligned}$$

This completes the proof. ■

In what follows, we denote the the Jacobian of $\left(x_m\left((Cx)_m - x^T C x\right)\right)_{m=1}^{3}$ by $J(x_1, x_2, x_3)$. The following propositions concern the stability of the rest points.

Proposition 5.2 Consider the rest point

$$(0,1,0)$$

For every $M \in \{3, 4, ...\}$ and for large enough T:

1. When $p > \frac{2}{3}$ the rest point is asymptotically stable.
2. When $p \leq \frac{2}{3}$ the rest point is unstable.

Proof. We find the eigenvalues of $J(0,1,0)$. They are:

$$\begin{bmatrix} -T & -3p+2 & -3pT+2T \end{bmatrix}$$

When $p > \frac{2}{3}$ (resp. $p < \frac{2}{3}$) all eigenvalues are negative (resp. the second and first eigenvalues are positive) and the rest point is asymptotically stable (resp. unstable). Now suppose $p = \frac{2}{3}$ and let us look at (5.1). With $p = \frac{2}{3}$ this reduces to

$$\frac{dx_1}{dt} = x_1 \left(\begin{bmatrix} MT & T & MT \end{bmatrix} x - x^T \begin{bmatrix} MT & T & MT \\ 2T & T & T+1 \\ MT & T & MT \end{bmatrix} x \right) \tag{5.6}$$

$$\frac{dx_2}{dt} = x_2 \left(\begin{bmatrix} 2T & T & T+1 \end{bmatrix} x - x^T \begin{bmatrix} MT & T & MT \\ 2T & T & T+1 \\ MT & T & MT \end{bmatrix} x \right) \tag{5.7}$$

We omit the third differential equation, since we know that $x_1(t) + x_2(t) + x_3(t) = 1$ for all t. It is easily checked that $(0,1)$ is a rest point of (5.6)-(5.7); now we will show that it is unstable. To this end it suffices to consider perturbations along the x_2 axis, i.e., with $x_1 = 0$. Hence, substituting $x_1 = 0$, $x_2 = 1 - \varepsilon$ and $x_3 = 1 - x_2 = \varepsilon$ in (5.6)-(5.7), we get

$$\begin{aligned}
\frac{dx_1}{dt} &= 0 \\
\frac{dx_2}{dt} &= -\frac{d\varepsilon}{dt} = -\left((M-1)T - 1\right)(1-\varepsilon)\,\varepsilon^2.
\end{aligned}$$

Clearly $\frac{dx_2}{dt} < 0$ and $x_2(t)$ will decrease away from 1. Hence $(0,1)$ is an unstable rest point of (5.6)-(5.7) and $(0,1,0)$ is an unstable rest point of (5.1). ■

Proposition 5.3 Consider the rest point

$$\left(\frac{3p-2}{M-2}, \frac{M-3p}{M-2}, 0\right)$$

For $p \geq \frac{2}{3}$ and for every $M \in \{3, 4, ...\}$ and large enough T the rest point is unstable.

Proof. We find the eigenvalues of $J\left(\frac{3p-2}{M-2}, \frac{M-3p}{M-2}, 0\right)$. They are:

$$\left[\; -\frac{T\left(9\,p^2+M-9pP\right)}{M-2} \quad \frac{3\,MpT-9\,p^2T-3\,Mp-2\,TM+9\,p^2+6pT+2\,M-6p}{M-2} \quad \frac{T\left(3\,Mp-9p^2-2\,M+6p\right)}{M-2} \;\right]$$

The rest point is admissible for $p \geq \frac{2}{3}$. For $p = \frac{2}{3}$ it becomes $(0,1,0)$ and we have already seen that it is unstable. Let us consider the case $p > \frac{2}{3}$. Conside the second eigenvalue:

$$\frac{3\,MpT - 9\,p^2T - 3\,Mp - 2\,TM + 9\,p^2 + 6pT + 2\,M - 6p}{M-2}.$$

The denominator $M - 2$ is always positive. The sign of the numerator is determined by the terms which are first-order in T. They are

$$3\,MpT - 9p^2T - 2\,TM + 6pT = T\,(3p-2)\,(M-3p) > 0$$

hence the rest point is unstable. ■

Proposition 5.4 Consider the rest point

$$\left(0, \frac{TM - T - 3p + 1}{TM - T - 1}, \frac{3p-2}{TM-T-1}\right),$$

For $p \geq \frac{2}{3}$ and for every $M \in \{3, 4, ...\}$ and large enough T the rest point is unstable.

Proof. The eigenvalues of $J\left(0, \frac{TM-T-3p+1}{TM-T-1}, \frac{3p-2}{TM-T-1}\right)$ are:

$$\begin{aligned}
&\frac{3\,MpT - 2\,TM - 9p^2 - 3pT + 9p + 2\,T - 2}{TM - T - 1}, \\
&-\frac{3\,MpT^2 - 3\,MpT - 2\,MT^2 - 9p^2T - 3\,T^2p + 2\,TM + 9p^2 + 12pT + 2\,T^2 - 9p - 4\,T + 2}{TM - T - 1}, \\
&-\frac{MT^2 + 9p^2 - T^2 - 9p - T + 2}{TM - T - 1}.
\end{aligned}$$

For the rest point to be admissible we must have $p \geq \frac{2}{3}$. For $p = \frac{2}{3}$ the rest point becomes $(0, 1, 0)$ which, as we have already seen is unstable. Let us consider the case $p > \frac{2}{3}$. The first eigenvalue is

$$\frac{3\,MpT - 2\,TM - 9p^2 - 3pT + 9p + 2\,T - 2}{TM - T - 1}$$

with positive denominator. The sign of the numerator is determined by the terms which are first-order in T. They are

$$3\,MpT - 2\,TM - 3pT + 2\,T = T\,(3p - 2)\,(M - 1) > 0.$$

Hence the rest pont is unstable. ■

Proposition 5.5 For every $M \in \{3, 4, ...\}$ and for large enough T, the set

$$\{(x_1, 0, 1 - x_1) : \text{ with } x_1 \in [0, 1]\}$$

is a stable non-isolated equilibrium set.

Proof. We find the eigenvalues of $J\,(x_1, 0, 1 - x_1)$. They are:

$$\begin{bmatrix} 0 & -TM & -3\,pTx_3 - TM + 3\,pT + 3\,px_3 + Tx_3 - x_3 \end{bmatrix}$$

The first eignevalue is zero and the second is negative.The sign of the third eigenvalue, for large T, is determined by

$$-3\,pTx_3 - TM + 3\,pT + Tx_3 = 3\,pT - 3\,pTx_3 - TM + Tx_3 = 3pT\,(1 - x_3) - T\,(M - x_3)$$

As the above is a linear function of p, its maximum value is achieved at $p = 1$ and we have

$$3pT\,(1 - x_3) - T\,(M - x_3) \leq 3T\,(1 - x_3) - T\,(M - x_3) = -T\,(M - 3 + 2x_3) \leq -T\,(M - 3) \leq 0.$$

All eigenvalues are nonpositive but, since at least one is zero, to determine stability we must look at the system (5.1) in more detail. Letting $x_1 = \varepsilon_1$, $x_2 = \varepsilon_2$, (5.1) reduces to

$$\frac{dx_1}{dt} = \frac{d\varepsilon_1}{dt} = T\,(M - 3p + 1)\,\varepsilon_1\varepsilon_2, \tag{5.8}$$

$$\frac{dx_2}{dt} = \frac{d\varepsilon_2}{dt} = -\,(TM - 1)\,\varepsilon_2. \tag{5.9}$$

If the system is perturbed from some $(a, 0, 1 - a)$ to some $(a + \varepsilon_1, \varepsilon_2, 1 - a - \varepsilon_1 - \varepsilon_2)$, we see from (5.8)-(5.9) that: the ε_2 perturbation will decrease to 0 while, at the same time, ε_1 will keep increasing until some time at which ε_2 is essentially zero. Consequently, the system will settle to a new rest point which will have the form $(a', 0, 1 - a')$, with $a' > a$. In short, while any rest point $(x_1, 0, 1 - x_1)$ is not asymptotically stable, the set $\{(x_1, 0, 1 - x_1) : \text{ with } x_1 \in [0, 1]\}$ is a non-isolated equilibrium set. ■

Example 5.6 We give two examples of the phase portait of (5.1). In Figure 2 we have $p = \frac{3}{4}$, $M = 4$, $T = 10$ and in Figure 3 we have $p = \frac{3}{4}$, $M = 4$, $T = 10$. In both cases we have the stable non-isolated equilibrium set $\{(x_1, 0, 1 - x_1) : \text{ with } x_1 \in [0, 1]\}$ and the rest point $(0, 1, 0)$ (stable for $= \frac{3}{4}$ and unstable for $p = \frac{3}{5}$). In the case $p = \frac{3}{4}$ we see that we also have the unstable rest points $\left(\frac{1}{8}, \frac{7}{8}, 0\right)$ and $\left(0, \frac{115}{116}, \frac{1}{116}\right)$.

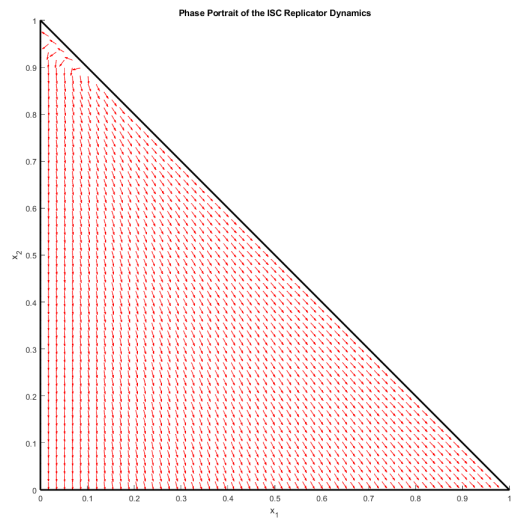

Figure 2: Phase portait of (5.1) for $p = 3/4$.

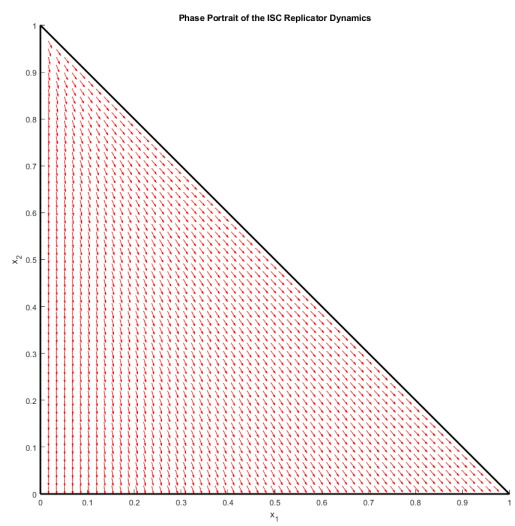

Figure 3: Phase portait of (5.1) for $p = 3/5$.

6 Evolutionary Stable Strategies

Recall the definition of *evolutionarily stable strategy* (ESS).

Definition 6.1 We say that an $\widehat{\mathbf{x}}$ is an *evolutionarily stable strategy* iff its satisfies the following two conditions

$$\forall \mathbf{x} : \widehat{\mathbf{x}}^T C \widehat{\mathbf{x}} \geq \mathbf{x}^T C \widehat{\mathbf{x}} \tag{6.1}$$

$$\forall \mathbf{x} \neq \widehat{\mathbf{x}} : \widehat{\mathbf{x}}^T C \widehat{\mathbf{x}} = \mathbf{x}^T C \widehat{\mathbf{x}} \Rightarrow \widehat{\mathbf{x}}^T C \mathbf{x} > \mathbf{x}^T C \mathbf{x}. \tag{6.2}$$

The following two propositions characterize the symmetric NE strategies in terms of evolutionary stability.

Proposition 6.2 For every $M \in \{3, 4, ...\}$ and for large enough T,

$$\left(\frac{3p-2}{M-2}, \frac{M-3p}{M-2}, 0\right) \text{ and } \left(\frac{3p-2}{M-2}, \frac{M-3p}{M-2}, 0\right)$$

are not ESS.

Proof. This is true because the rest points $\left(\frac{3p-2}{M-2}, \frac{M-3p}{M-2}, 0\right)$ and $\left(\frac{3p-2}{M-2}, \frac{M-3p}{M-2}, 0\right)$ are asymptotically unstable [6]. ■

Proposition 6.3 For every $M \in \{3, 4, ...\}$ and for large enough T, $(0, 1, 0)$ is not ESS (is ESS) when $p \leq \frac{2}{3}$ (when $p > \frac{2}{3}$).

Proof. When $p \leq \frac{2}{3}$ is an unstable rest point, so $((0, 1, 0), (0, 1, 0))$ cannot be ESS. Let us consider the case $p > \frac{2}{3}$. Consider the first condition

$$\forall \mathbf{x} : \widehat{\mathbf{x}}^T C \widehat{\mathbf{x}} \geq \mathbf{x}^T C \widehat{\mathbf{x}}.$$

Take any $\mathbf{x} = (a, b, 1 - a - b)$. We have

$$\widehat{\mathbf{x}}^T C \widehat{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} TM & 3(1-p)T & TM \\ 3pT & T & 3p + (T-1) \\ TM & 3(1-p) + (T-1) & TM \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = T$$

$$\begin{aligned} \mathbf{x}^T C \widehat{\mathbf{x}} &= \begin{bmatrix} a & b & 1-a-b \end{bmatrix} \begin{bmatrix} TM & 3(1-p)T & TM \\ 3pT & T & 3p + (T-1) \\ TM & 3(1-p) + (T-1) & TM \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \\ &= 2aT - 3aTp + 2 - 3p + T - 2a + 3ap - 2b + 3bp. \end{aligned}$$

Then, for laarge T,

$$\widehat{\mathbf{x}}^T C \widehat{\mathbf{x}} - \mathbf{x}^T C \widehat{\mathbf{x}} \simeq -2aT + 3aTp = Ta(3p - 2) > 0$$

since $p > \frac{2}{3}$. Hence the condition

$$\forall \mathbf{x} \neq \widehat{\mathbf{x}} : \widehat{\mathbf{x}}^T C \widehat{\mathbf{x}} = \mathbf{x}^T C \widehat{\mathbf{x}} \Rightarrow \widehat{\mathbf{x}}^T C \mathbf{x} > \mathbf{x}^T C \mathbf{x}.$$

need not be checked and we conclude that $(0, 1, 0)$ is ESS for $p > \frac{2}{3}$. ■

Proposition 6.4 For $M \in \{4, ...\}$ and for large enough T, the set of strategies

$$\mathcal{A} = \{(a, 0, 1-a) : a \in [0,1]\}$$

are an ESS set.

Proof. Take $\widehat{\mathbf{x}} = (a, 0, 1-a)$ and $\mathbf{x} = (b, c, 1-c)$. Let us first check the condition (6.1). We have

$$\widehat{\mathbf{x}}^T C\widehat{\mathbf{x}} = \begin{bmatrix} a & 0 & 1-a \end{bmatrix} \begin{bmatrix} TM & 3(1-p)T & TM \\ 3pT & T & 3p+(T-1) \\ TM & 3(1-p)+(T-1) & TM \end{bmatrix} \begin{bmatrix} a \\ 0 \\ 1-a \end{bmatrix} = TM$$

$$\begin{aligned} \mathbf{x}^T C\widehat{\mathbf{x}} &= \begin{bmatrix} b & c & 1-b-c \end{bmatrix} \begin{bmatrix} TM & 3(1-p)T & TM \\ 3pT & T & 3p+(T-1) \\ TM & 3(1-p)+(T-1) & TM \end{bmatrix} \begin{bmatrix} a \\ 0 \\ 1-a \end{bmatrix} \\ &= 3acpT + 3cp - 3cpa + cT - cTa - c + ca + TM - TMc \end{aligned}$$

Consequently

$$\begin{aligned} \widehat{\mathbf{x}}^T C\widehat{\mathbf{x}} - \mathbf{x}^T C\widehat{\mathbf{x}} &= TM - (3acpT + 3cp - 3cpa + cT - cTa - c + ca + TM - TMc) \\ &= -3acpT - 3cp + 3cpa - cT + cTa + c - ca + TMc \end{aligned}$$

For large enough T, the sign of $\widehat{\mathbf{x}}^T C\widehat{\mathbf{x}} - \mathbf{x}^T C\widehat{\mathbf{x}}$ is determined by the terms which are first-order in T, namely:

$$\begin{aligned} -3acpT - cT + cTa + TMc &= Tc(M - 1 + a(1-3p)) \\ &\geq Tc(M - 1 + 1(1-3)) = Tc(M-3) \geq 0 \end{aligned}$$

Since we have assumed $M \geq 4$, we will have $Tc(M-3) = 0$ only when $c = 0$. Hence (6.1) is satisfied. Regarding (6.2) we have

$$0 = \widehat{\mathbf{x}}^T C\widehat{\mathbf{x}} - \mathbf{x}^T C\widehat{\mathbf{x}} \geq Tc(M-3) \geq 0 \Rightarrow c = 0.$$

Conversely, for every $\mathbf{x} = (b, c, 1-b-c)$ with $c > 0$, we will have

$$\widehat{\mathbf{x}}^T C\widehat{\mathbf{x}} > \mathbf{x}^T C\widehat{\mathbf{x}}.$$

Hence any strategy $\widehat{\mathbf{x}} = (a, 0, 1-a)$ can only be invaded by strategies $\mathbf{x} = (b, 0, 1-b)$; in this sense the set $\mathcal{A}$ is an evolutionarily stable set. ■

7 Markovian Analysis

7.1 Markov Chains

Before embarking in the Markovian analysis of the ISC, we present a brief overview of the relevant concepts regarding Markov chains. For details see [9].

Definition 7.1 A *Markov chain* (MC) is a stochastic process $(U_t)_{t=0}^{\infty}$ taking values in a finite set $\mathcal{U} = \{u_1, ..., u_K\}$ (called the *state space*) and satisfying the *Markovian property*:

$$\forall t, n \in \{0, 1, ...\} : \Pr\left(U_{t+1} = u_{t+1} | U_0 = u_0, U_1 = u_1, ..., U_t = u_t\right) = \Pr\left(U_{t+1} = u_{t+1} | U_t = u_t\right).$$

When $\Pr\left(U_{t+1} = u_{t+1} | U_t = u_t\right)$ is independent of t, we say that the MC is *homogeneous* and the *transition matrix* of $(U_t)_{t=0}^{\infty}$ is the $K \times K$ matrix P defined by

$$\forall u, u' \in \mathcal{U} : P_{u,u'} = \Pr\left(U_{t+1} = u' | U_t = u\right).$$

The members of $\mathcal{U}$ are called the *states* of $(U_t)_{t=0}^{\infty}$ and can be characterized as follows [9].

1. A *transient* state is one which, with probability one, will be visited a finite number of times.
2. A *persistent* state is one which, with probability one, will be visited infinitely often.
3. An *absorbing* state is one which, with probability one, always transitions to itself.

Furthermore, we say that two states u, u' *communicate* with each other (we write $u \leftrightharpoons u'$) if each is reachable from the other with positive probability. The communication relation $\leftrightharpoons$ is an equivalence relation, and we call each of its equivalence classes a *communicating class*. A communicating class is *closed* if the probability of leaving the class is zero.

7.2 Evolutionary Games and Markov Chains

We have previously assumed that, in an evolutionary game, the population evolves according to the replicator dynamics. In this section we will justify this assumption. The first step is to define evolutionary games. We start with a *symmetric* two-player game described by an $M \times M$ bimatrix $\left(A, A^T\right)$. We assume the existence of N *players* and define the following:

1. A *population* vector $\mathbf{z} = (z_1, ..., z_N)$ belonging to

   $$\mathcal{Z} = \{(z_1, ..., z_N) : \forall n : z_n \in \{1, .., M\}\}.$$

   This indicates the strategy used by each player: $z_n = m$ means the n-th player is using the m-th strategy; we will often call such a player an "*m-user*".

2. A *strategy* or *state* vector $\mathbf{s} = (s_1, ..., s_M) \in \mathcal{S}_M$ indicates how many players are using each strategy, i.e., s_m is the number of m-users. It takes values in the *State Space*:

   $$\mathcal{S}_M = \left\{(s_1, ..., s_M) \in \{0, 1, .., N\}^M \text{ s.t. } \sum_{m=1}^{M} s_m = N\right\}.$$

   We also define the following subsets of $\mathcal{S}_M$:

   $$\textit{Tips} : \mathcal{S}_M^{(t)} = \{(1, 0, 0, ..., 0, 0), (0, 1, 0, ..., 0, 0), ..., (0, 0, 0, ..., 0, 1)\}$$

$$\textit{Interior}: \mathcal{S}_M^{(i)} = \left\{ (s_1, ..., s_M) \in \{1, .., N-1\}^M \text{ s.t. } \sum_{m=1}^{M} s_m = N \right\}$$

$$\textit{Face}: \mathcal{S}_M^{(f,m)} = \left\{ (s_1, ..., s_M) \in \{1, .., N-1\}^M \text{ s.t. } \sum_{m=1}^{M} s_m = N \text{ and } s_m = 0 \right\}$$

Clearly, $\left(\mathcal{S}_M^{(t)}, \mathcal{S}_M^{(i)}, \mathcal{S}_M^{(f,1)}, ..., \mathcal{S}_M^{(f,M)}\right)$ is a partition of $\mathcal{S}_M$.

3. A *frequency* vector $\mathbf{x} = (x_1, ..., x_M)$, where for each $m \in \{1, ..., M\}$ we define $x_m = \frac{s_m}{N}$. Hence x_m indicates the proportion of m-users. $\mathbf{x}$ takes values in Δ_M.

For given payoff matrix A, and number of players N, the evolutionary game consists of the following procedure.

Evolutionary Game Protocol

1. Choose an initial distribution of strategies to players: $\mathbf{z}(0) = (z_1(0), ..., z_N(0)) \in \mathbf{Z}$.
2. Repeat for $j \in \{0, 1, ..., J-1\}$ the following.
   (a) Compute $\mathbf{s}(j) = (s_1(j), ..., s_M(j)) \in \mathbf{S}$ and $\mathbf{x}(j) = (x_1(j), ..., x_M(j))$:
   $$\forall m : s_m(j) = |\{z_n(j) : z_n(j) = m\}|, \quad x_m(j) = \frac{s_m(j)}{\sum_{m=1}^{M} s_m(j)}.$$
   (b) Play all possible matches between players, i.e., for all the pairs (n_1, n_2) with $n_1, n_2 \in \{1, ..., N\}$ and $n_1 \neq n_2$. In each match, n_1 and n_2 play one round of the A game.
   (c) The n-th player (for $n \in \{1, ..., N\}$) collects a total payoff
   $$q_n(z(j)) = \sum_{n \neq n'} A_{z_n(j) z_{n'}(j)}.$$
   (d) The m-th strategy collects a total payoff
   $$Q_m(s(j)) = \sum_{n : z_n = m} q_n(z(j)).$$
   (e) Obtain the next population vector $\mathbf{z}(j+1)$ (also called a *generation*) by letting one player update his strategy, by a *revision protocol* $\mathcal{R}$.

So an evolutionary game is a pair $(A, \mathcal{R})$ (with auxiliary parameters N and J) and produces sequences $(\mathbf{z}(j))_{j=0}^{\infty}$, $(\mathbf{s}(j))_{j=0}^{\infty}$ and $(\mathbf{x}(j))_{j=0}^{\infty}$. It is clear that the population process $(\mathbf{z}(j))_{j=0}^{\infty}$

is a Markov chain and it is easy to show that the same is true of the state process $(\mathbf{s}(j))_{j=0}^{\infty}$; the *frequency process* $(\mathbf{x}(j))_{j=0}^{\infty}$ is simply a function of $(\mathbf{s}(j))_{j=0}^{\infty}$.

To complete the description of an evolutionary game we must define a *revision protocol*. We will only consider the *pairwise proportional imitation* protocol. To define this we need some preliminary definitions.

Definition 7.2 We will write the player total payoff vector as $\mathbf{q}(\mathbf{z}) = (q_1(\mathbf{z}), ..., q_N(\mathbf{z}))$ and the strategy total payoff vector as $\mathbf{Q} = (Q_1(\mathbf{s}), ..., Q_M(\mathbf{s}))$. Also we use the notation $[y]_+ = \max(y, 0)$.

Definition 7.3 The *Pairwise Proportional Imitation* (PPI) revision protocol is defined as follows.

1. A player is selected equiprobably from the population.
2. If the player is an m_1-user, then:
   (a) If
   $$Q_{m_1}(\mathbf{s}) = \max_m Q_m(\mathbf{s})$$
   then the player keeps the m_1-th strategy in the next iteration.
   (b) Otherwise, the player adopts the m_2-th strategy with probability $\rho_{m_1 m_2}(\mathbf{s})$, where
   $$\rho_{m_1 m_2}(\mathbf{s}) = \frac{x_{m_2}[Q_{m_2}(\mathbf{s}) - Q_{m_1}(\mathbf{s})]_+}{\sum_{m=1}^{M} x_m [Q_m(\mathbf{s}) - Q_{m_1}(\mathbf{s})]_+}.$$
   In other words, $\rho_{m_1 m_2}(\mathbf{s})$ is proportional to the positive part of the surplus of Q_{m_2} over Q_{m_1}, multiplied by the frequency of m_2.

The following theorem [16] shows that the PPI revision protocol corresponds to the replicator dynamics.

Proposition 7.4 If in the evolutionary game $(A, \mathcal{R})$ the revision protocol is PPI then, in the limit $N \to \infty$, the $\mathbf{x}(t)$ process follows the *Replicator Dynamics*:

$$\forall m : \frac{dx_m}{dt} = x_m \left(\mathbf{e}_m^T A\mathbf{x} - \mathbf{x}^T A\mathbf{x}\right).$$

7.3 Absorbing States and Basins of Attraction

We will now characterize the states of the ISC Markov chain, i.e., we will identify the transient, persistent and absorbing states, as well as the ergodic classes. Recall that, by the PPI rule, the selected i-user:

1. stays to the i-th strategy with probability one iff $[Q_j - Q_i]_+ = 0$ for all $j \neq i$,

2. otherwise he imitates a j-user (with $j \neq i$) with probability

$$\rho_{ij}(s_1, s_2, s_3) \sim x_j \left[Q_j(s_1, s_2, s_3) - Q_i(s_1, s_2, s_3)\right]_+ .$$

The ρ_{ij} 's are used to compute the state transition probabilities $P_{\mathbf{s},\mathbf{s}'}$. To characterize the states of ISC as transient or persistent we do not need to know the actual $P_{\mathbf{s},\mathbf{s}'}$ values; we only need to know, for each $(\mathbf{s}, \mathbf{s}')$ pair, whether $P_{\mathbf{s},\mathbf{s}'}$ is positive or zero. We will now show how this can be achieved using the ρ_{ij}'s.

Let us first look at the interior states, i.e., the $\mathbf{s} = (s_1, s_2, s_3)$ which satisfy $s_1, s_2, s_3 > 0$. The total payoff at $\mathbf{s} = (s_1, s_2, s_3)$ is

$$Q(s_1, s_2, s_3) = \begin{bmatrix} (s_1 - 1)TM + 3s_2(1-p)T + (N - s_1 - s_2)TM \\ \\ 3s_1pT + (s_2 - 1)T + (N - s_1 - s_2)(3p + T - 1) \\ \\ s_1TM + s_2(2 - 3p + T) + (N - s_1 - s_2 - 1)TM \end{bmatrix}.$$

We define a 3×3 matrix $R(s_1, s_2, s_3)$ as follows:

$$\begin{aligned} &\forall i \neq j : R_{ij}(s_1, s_2, s_3) = \left[Q_j(s_1, s_2, s_3) - Q_i(s_1, s_2, s_3)\right]_+ , \\ &\forall i : R_{ii}(s_1, s_2, s_3) = \begin{cases} 0 & \text{iff } \max_{j \neq i} Q_j(s_1, s_2, s_3) > Q_i(s_1, s_2, s_3) \\ 1 & \text{iff } \max_{j \neq i} Q_j(s_1, s_2, s_3) \leq Q_i(s_1, s_2, s_3) \end{cases} . \end{aligned}$$

We see that

$$\forall (s_1, s_2, s_3) \in S_3^i : R_{ij}(s_1, s_2, s_3) > 0 \Leftrightarrow \rho_{ij}(s_1, s_2, s_3) > 0.$$

Hence, to locate the positive elements of ρ it suffices to locate the positive elements of R. These are computed from the payoff vector $(\mathbf{Q}(\mathbf{s}))_{\mathbf{s} \in S}$. The computation is tedious[1] and we omit the results. The important point is that each $R_{ij}(\mathbf{s})$ has a term $\overline{R}_{ij}(\mathbf{s})$ which is first-order in T and a term $c_{ij}(\mathbf{s})$ which is independent of T:

$$\forall i, j, \mathbf{s} : R_{ij}(\mathbf{s}) = \overline{R}_{ij}(\mathbf{s}) + c_{ij}(\mathbf{s}).$$

For large T, $\overline{R}_{ij}(\mathbf{s})$ dominates $c_{ij}(\mathbf{s})$ and determines the sign $R_{ij}(\mathbf{s})$. Once again, omitting computations, let us list the $\overline{R}_{ij}(\mathbf{s})$ terms:

$$\overline{R}_{12}(\mathbf{s}) = Ts_2\left((3p-1)s_1 + (M + 3p - 3)s_2 - (M-1)(N-1)\right), \tag{7.1}$$

$$\overline{R}_{13}(\mathbf{s}) = Ts_2s_3(3p-2), \tag{7.2}$$

$$\overline{R}_{21}(\mathbf{s}) = -Ts_1\left((3p-1)s_1 + (M + 3p - 3)s_2 - (M-1)(N-1)\right), \tag{7.3}$$

$$\overline{R}_{23}(\mathbf{s}) = T(N - s_1 + s_2)\left((1-3p)s_1 + (1-M)s_2 + (1-M)(1-N)\right), \tag{7.4}$$

$$\overline{R}_{31}(\mathbf{s}) = -Ts_1s_2(3p-2), \tag{7.5}$$

$$\overline{R}_{32}(\mathbf{s}) = -Ts_2\left((1-3p)s_1 + (1-M)s_2 + (1-N)(1-M)\right). \tag{7.6}$$

[1] It was performed by Maple.

We will use $\overline{R}$ to characterize state transitions. In the following discussion we exclude "self-transitions" of the form:

$$(s_1, s_2, s_3) \to (s_1, s_2, s_3)$$

since these do not influence the persistence or transience of (s_1, s_2, s_3). The following propositions characterize transitions from states belonging to specific subsets of $\mathcal{S}_3$.

The first proposition concerns interior states.

Proposition 7.5 Assume $M, N \in \{4, ...\}$ and T is large enough. For any $(s_1, s_2, s_3) \in \mathcal{S}_3^{(i)}$, excluding self-transitions, the only transitions of positive probability are:

$$\begin{aligned}
\text{south:}\ & (s_1, s_2, s_3) \to (s_1, s_2 - 1, s_3 + 1)\,, \\
\text{southeast:}\ & (s_1, s_2, s_3) \to (s_1 + 1, s_2 - 1, s_3)\,, \\
\text{west:}\ & (s_1, s_2, s_3) \to (s_1 - 1, s_2, s_3 + 1)\ \text{ iff } p > \frac{2}{3}, \\
\text{east:}\ & (s_1, s_2, s_3) \to (s_1 + 1, s_2, s_3 - 1)\ \text{ iff } p \leq \frac{2}{3}.
\end{aligned}$$

Proof. The key observation is that, for interior states (s_1, s_2, s_3), PPI dictates that transitions of positive probability can be characterized as follows:

$$\forall\,(s_1, s_2, s_3) \in \mathcal{S}_3^{(i)} : P_{(s_1,s_2,s_3),(s_1',s_2',s_3')} > 0 \Leftrightarrow R_{ij}\,(s_1, s_2, s_3) > 0 \Leftrightarrow \rho_{ij}\,(s_1, s_2, s_3) > 0.$$

Hence (7.1)-(7.6) determine the transitions of positive probability. We give an example for transitions of the form

$$(s_1, s_2, s_3) \to (s_1 - 1, s_2 + 1, s_3)\ \text{ and }\ (s_1, s_2, s_3) \to (s_1 + 1, s_2 - 1, s_3)$$

("northwest" and "southeast" transitions). For the transition $(s_1, s_2, s_3) \to (s_1 - 1, s_2 + 1, s_3)$, a 1-user must be selected (this has probability $\frac{1}{3}$) and $\overline{R}_{12}\,(s_1, s_2, s_3)$ must be positive. From (7.1) we know that

$$\overline{R}_{12}\,(s_1, s_2, s_3) = T s_2\,((3p - 1)\,s_1 + (M + 3p - 3)\,s_2 - (M - 1)\,(N - 1))\,.$$

Hence

$$\Pr\left((s_1, s_2, s_3) \to (s_1 - 1, s_2, s_3 + 1)\ \right) > 0 \Leftrightarrow (3p - 1)\,s_1 + (M + 3p - 3)\,s_2 - (M - 1)\,(N - 1) > 0.$$

Now let us apply this reasoning to (7.1)-(7.6), separated into three groups.

1. From (7.4) and (7.6) we see that

$$\overline{R}_{23}\,(\mathbf{s}) > 0 \Leftrightarrow (1 - 3p)s_1 + (1 - M)s_2 + (1 - M)\,(1 - N) > 0, \tag{7.7}$$

$$\overline{R}_{32}\,(\mathbf{s}) > 0 \Leftrightarrow (1 - 3p)s_1 + (1 - M)s_2 + (1 - M)\,(1 - N) < 0. \tag{7.8}$$

The line

$$l_2 : (1 - 3p)s_1 + (1 - M)s_2 + (1 - M)\,(1 - N) = 0$$

passes through $\left(\frac{(M-1)(N-1)}{1-3p}, 0\right)$ and $(0, N-1)$. Since

$$\frac{(M-1)(N-1)}{1-3p} \leq -\frac{(M-1)(N-1)}{2}$$

the entire $\mathcal{S}_3^{(i)}$ is in one of the half-planes and, since

$$(1-3p)0 + (1-M)0 + (1-M)(1-N) = (1-M)(1-N) > 0$$

it is in the positive half-plane. Hence we always have $\overline{R}_{23}(\mathbf{s}) > 0$ and $\overline{R}_{32}(\mathbf{s}) < 0$ and we conclude that

$$\begin{aligned}
&\Pr((s_1, s_2, s_3) \to (s_1, s_2 - 1, s_3 + 1)) > 0, \\
&\Pr((s_1, s_2, s_3) \to (s_1, s_2 + 1, s_3 - 1)) = 0,
\end{aligned}$$

i.e., only the south transitions have positive probability.

2. From (7.1) and (7.3) we see that

$$\begin{aligned}
&(s_1, s_2, s_3) \to (s_1 + 1, s_2 - 1, s_3) \Leftrightarrow (3p-1)s_1 + (M + 3p - 3)s_2 - (M-1)(N-1) > 0, \\
&(s_1, s_2, s_3) \to (s_1 - 1, s_2 + 1, s_3) \Leftrightarrow (3p-1)s_1 + (M + 3p - 3)s_2 - (M-1)(N-1) < 0.
\end{aligned}$$

The line

$$l_1 : ((3p-1)s_1 + (M + 3p - 3)s_2 - (M-1)(N-1)) = 0$$

intersects the s_1 and s_2 axes at the points $\left(\frac{(M-1)(N-1)}{3p-1}, 0\right)$ and $\left(0, \frac{(M-1)(N-1)}{M+3p-3}\right)$, respectively. Since

$$\max_{p \in \left[\frac{1}{2}, 1\right]} (3p-1) = 2$$

it follows that

$$\min_{p \in \left[\frac{1}{2}, 1\right]} \frac{(M-1)(N-1)}{3p-1} = \frac{(M-1)(N-1)}{2} > N.$$

Also

$$\begin{aligned}
(3p-1)s_1 + (M + 3p - 3)s_2 - (M-1)(N-1) &= 3pN - 3p - 2 + 2M - MN \\
&< \frac{3}{2}N - 3 - 2 + 2M - MN \\
&= \left(\frac{3}{2} - M\right)N + 2M \\
&< \left(\frac{3}{2} - M\right)4 + 2M = 6 - 2M < 0.
\end{aligned}$$

Hence the entire $\mathcal{S}_3^{(i)}$ lies below l_1 and

$$\begin{aligned}
&\Pr((s_1, s_2, s_3) \to (s_1 + 1, s_2 - 1, s_3)) > 0, \\
&\Pr((s_1, s_2, s_3) \to (s_1 - 1, s_2 + 1, s_3)) = 0.
\end{aligned}$$

i.e., only the southeast transitions have positive probability.

3. From (7.2) and (7.5) we see that the following transitions have positive probability:

$$\text{when } p > \frac{2}{3} : (s_1, s_2, s_3) \to (s_1 - 1, s_2, s_3 + 1),$$
$$\text{when } p \leq \frac{2}{3} : (s_1, s_2, s_3) \to (s_1 + 1, s_2, s_3 - 1).$$

I.e., for $p > \frac{2}{3}$ (resp. $p \leq \frac{2}{3}$) only west (resp. east) transitions have positive probability.

This completes the proof. ■

The following propositions concern the face states.

Proposition 7.6 Assume $M, N \in \{4, ...\}$, and T large enough. Then, for any $(0, s_2, s_3) \in \mathcal{S}_3^{(1)} \backslash \{(0, 0, N), (0, N, 0), (0, N-1, 0)\}$, excluding self-transitions, the only transitions of positive probability is:

$$\text{south}: (s_1, s_2, s_3) \to (s_1, s_2 - 1, s_3 + 1).$$

Proof. Let us examine the signs of $\overline{R}_{23}(s_1, s_2, s_3)$ and $\overline{R}_{32}(s_1, s_2, s_3)$. From (7.4) we see that

$$\overline{R}_{23}(s_1, s_2, s_3) > 0 \Leftrightarrow T(N - s_1 + s_2)((1 - 3p)s_1 + (1 - M)s_2 + (1 - M)(1 - N)) > 0.$$

Since $s_1 = 0$, we need

$$T(N + s_2)((1 - M)s_2 + (1 - M)(1 - N)) > 0 \Leftrightarrow$$
$$((1 - M)s_2 + (1 - M)(1 - N)) = (N - s_2 - 1)(M - 1) > 0$$

which will be true (since $s_2 < N - 1$). Similarly, from (7.6) we see that

$$\overline{R}_{32}(s_1, s_2, s_3) > 0 \Leftrightarrow (1 - 3p)s_1 + (1 - M)s_2 + (1 - N)(1 - M) < 0$$

for which we only need

$$(1 - M)s_2 + (1 - N)(1 - M) = (M - 1)(N - 1 - s_2) < 0$$

but this will never be true. Hence we have

$$\Pr((s_1, s_2, s_3) \to (s_1, s_2 - 1, s_3 + 1)) > 0,$$
$$\Pr((s_1, s_2, s_3) \to (s_1, s_2 + 1, s_3 - 1)) = 0.$$

This completes the proof. ■

Proposition 7.7 Assume $M, N \in \{4, ...\}$, and T large enough. Then, for any $(s_1, 0, s_3) \in \mathcal{S}_3^{(2)}$ the only transitions of positive probability are the self-transitions.

Proof. Since $s_2 = 0$ on $\mathcal{S}_3^{(2)}$, we will never have transitions

$$(s_1, 0, s_3) \to (s_1', 1, s_3').$$

So we only need to consider the transitions

$$(s_1, 0, s_3) \to (s_1 - 1, 0, s_3 + 1),$$
$$(s_1, 0, s_3) \to (s_1 + 1, 0, s_3 - 1).$$

We have

$$\overline{R}_{13}(s_1, 0, s_3) = T0s_3(3p - 2) = 0,$$
$$\overline{R}_{31}(s_1, 0, s_3) = -Ts_10(3p - 2) = 0.$$

So these transitions are also impossible. ■

Proposition 7.8 Assume $M, N \in \{4, ...\}$, and T large enough. Then, for any $(s_1, s_2, 0) \in \mathcal{S}_3^{(3)}$, excluding self-transitions, the only transitions of positive probability are:

northwest: when $(3p - 1)s_1 + (M + 3p - 3)s_2 - (M - 1)(N - 1) > 0 : (s_1, s_2, 0) \to (s_1 - 1, s_2 + 1, 0)$,
southeast: when $(3p - 1)s_1 + (M + 3p - 3)s_2 - (M - 1)(N - 1) < 0 : (s_1, s_2, 0) \to (s_1 + 1, s_2 - 1, 0)$.

Proof. We only need to consider transitions of the form

$$(s_1, s_2, 0) \to (s_1 - 1, s_2 + 1, 0),$$
$$(s_1, s_2, 0) \to (s_1 + 1, s_2 - 1, 0).$$

We have

$$\overline{R}_{12}(s_1, s_2, 0) = Ts_2((3p - 1)s_1 + (M + 3p - 3)s_2 - (M - 1)(N - 1)),$$
$$\overline{R}_{21}(s_1, s_2, 0) = -Ts_1((3p - 1)s_1 + (M + 3p - 3)s_2 - (M - 1)(N - 1)).$$

The sign in both cases is determined by

$$((3p - 1)s_1 + (M + 3p - 3)s_2 - (M - 1)(N - 1)),$$

namely

$$(3p - 1)s_1 + (M + 3p - 3)s_2 - (M - 1)(N - 1) > 0 \Rightarrow \Pr((s_1, s_2, 0) \to (s_1 - 1, s_2 + 1, 0)) > 0,$$
$$(3p - 1)s_1 + (M + 3p - 3)s_2 - (M - 1)(N - 1) < 0 \Rightarrow \Pr((s_1, s_2, 0) \to (s_1 + 1, s_2 - 1, 0)) > 0.$$

This completes the proof. ■

A geometric interpretation is possible. Consider the line

$$l_1 : (3p - 1)s_1 + (M + 3p - 3)s_2 = (M - 1)(N - 1).$$

It intersects the s_1 axis at $s_1 = \frac{(M-1)(N-1)}{3p-1}$ and the s_2 axis at $s_2 = \frac{(M-1)(N-1)}{-3+3p+M}$. For points (s_1, s_2):

$$\begin{aligned} &\text{below } l_2 : (3p-1)\, s_1 + (M+3p-3)\, s_2 - (M-1)\,(N-1) < 0, \\ &\text{above } l_2 : (3p-1)\, s_1 + (M+3p-3)\, s_2 - (M-1)\,(N-1) > 0. \end{aligned}$$

Hence for states (s_1, s_2, s_3):

$$\begin{aligned} &\text{below } l_2 : \Pr\left((s_1, s_2, 0) \to (s_1 - 1, s_2 + 1, 0)\right) = 0,\ \Pr\left((s_1, s_2, 0) \to (s_1 + 1, s_2 - 1, 0)\right) > 0, \\ &\text{above } l_2 : \Pr\left((s_1, s_2, 0) \to (s_1 - 1, s_2 + 1, 0)\right) > 0,\ \Pr\left((s_1, s_2, 0) \to (s_1 + 1, s_2 - 1, 0)\right) = 0. \end{aligned}$$

Finally, the next proposition characterizes the absorbing states and (partly) the basins of attraction.

Proposition 7.9 The absorbing states of the ISC are $(0, N, 0)$ and all $(s_1, s_2, s_3) \in S_3^{(2)}$, i.e., all the states of the form

$$\forall n \in \{0, 1, ..., N\} : (n, 0, N - n)\,.$$

Furthermore:

1. The states $(0, N-2, 2)\,,\ (0, N-3, 3), ..., (0, 1, N-1)$ are eventually absorbed into $(0, 0, N)$.
2. The state $(0, N-1, 1)$ is eventually absorbed into $(0, N, 0)$.
3. The states $(s_1, s_2, 0)$ are eventually absorbed:

$$\begin{aligned} &\text{into } (0, N, 0) \text{ when } (3p-1)\, s_1 + (M+3p-3)\, s_2 - (M-1)\,(N-1) > 0, \\ &\text{into } (N, 0, 0) \text{ when } (3p-1)\, s_1 + (M+3p-3)\, s_2 - (M-1)\,(N-1) < 0. \end{aligned}$$

4. The states $(s_1, s_2, s_3) \in S_3^{(i)}$ are eventually absorbed into some $(s_1, s_2, s_3) \in S_3^{(f,2)} \cup \{(N, 0, 0)\,, (0, 0, N)\}$.

Proof. It is obvious that the pure states $(N, 0, 0)$, $(0, N, 0)$ and $(0, 0, N)$ are absorbing. The states $(s_1, s_2, s_3) \in S_3^{(2)}$ are absorbing by Proposition 7.7. Regarding the transient states, we will examine the following cases in the order listed above.

1. The states $(0, N-2, 2)\,,\ (0, N-3, 3), ..., (0, 1, N-1)$ can only perform south transitions, from Proposition 7.7; hence eventually are absorbed into $(0, 0, N)$.
2. The state $(0, N-1, 1)$ can only perform a north transitions, so it is eventually absorbed into $(0, N, 0)$.
3. The states $(s_1, s_2, 0)$, from Proposition 7.8, depending on $(3p-1)\, s_1 + (M+3p-3)\, s_2 - (M-1)\,(N-1)$, can only perform either northwest transitions (in which case they are absorbed into $(0, N, 0)$) or southeast transitions (in which case they are absorbed into $(N, 0, 0)$).
4. The intrerior states $(s_1, s_2, s_3) \in S_3^{(i)}$, from Proposition 7.5, can only perform south, southeast, west and east transitions. Hence they always transit to (s_1', s_2', s_3') with $s_2' \leq s_2$ and eventually will reach some state $(s_1', 0, s_3')$ which is absorbing.

This completes the proof. ■

The results of the proposition are illustrated in the following Figure 3 (for $p = \frac{3}{4} > \frac{2}{3}$) and Figure 4 (for $p = \frac{3}{5} < \frac{2}{3}$). These figures are obtained when the number of players is $N = 10$. The plot is generated as follows. First, recall that every state (s_1, s_2, s_3) satisfies $s_1 + s_2 + s_3 = N$. Hence, given N and knowing two of s_1, s_2, s_3 uniquely specifies the third one. Then we can map every state (s_1, s_2, s_3) to a unique point $W(s_1, s_2, s_3) = (s_1, s_2)$ in the $(s_1$-$s_2)$-simplex. A nonzero element $P_{(s_1,s_2,s_3),(s_1',s_2',s_3')} > 0$ indicates a possible transition from (s_1, s_2, s_3) to (s_1', s_2', s_3'), which we indicate by an arrow from (s_1, s_2) to (s_1', s_2'); we omit self-loops. Also, since the states are arranged on an orthogonal grid, given a state (s_1, s_2, s_3) with $W(s_1, s_2, s_3) = (s_1, s_2)$, we can refer to the state (s_1', s_2', s_3') such that $W(s_1', s_2', s_3') = (s_1, s_2 + 1)$ as the "north neighbor" of (s_1, s_2, s_3), to (s_1'', s_2'', s_3'') such that $W(s_1'', s_2'', s_3'') = (s_1 + 1, s_2 - 1)$ as its "southeast neighbor" etc. These are *geometric* neighbors; if they are also neighbors in the graph theoretic sense (i.e., connected by an edge), then we say they are *connected* neighbors.

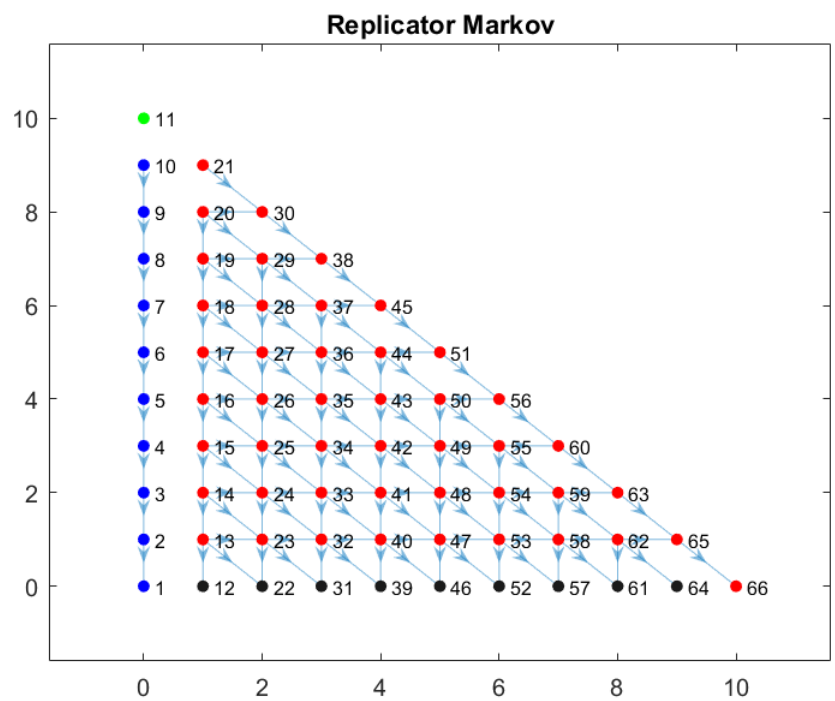

Figure 4: State transition graph for $p = 3/4$.

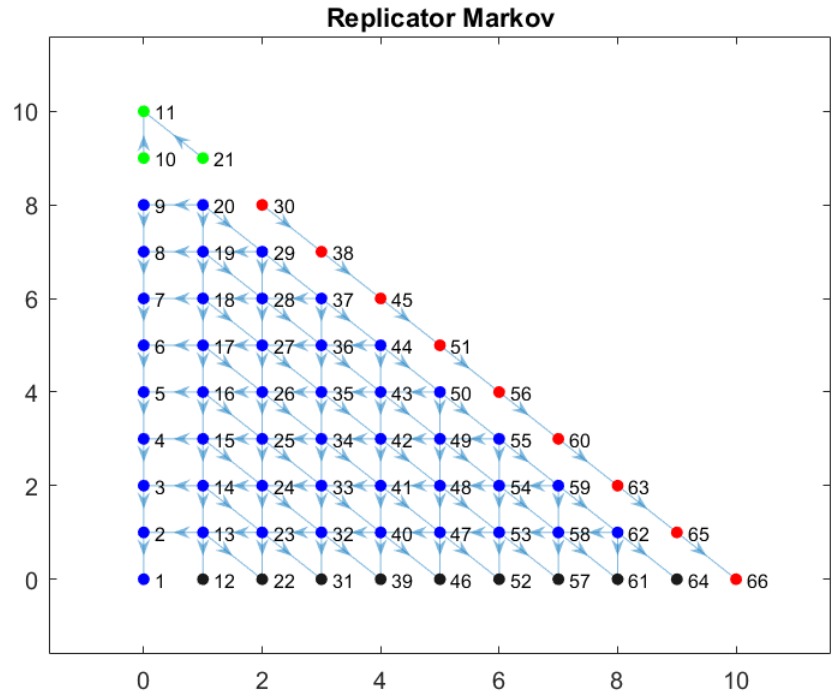

Figure 5: State transition graph for $p = 3/5$.

8 Connection to Learning Theory

Both evolutionary game theory (EGT) and learning models in games describe adaptive behavior in strategic environments; they are using identical dynamics but admit different interpretations, as we will now explain.

In EGT, a large population repeatedly plays a game and strategies reproduce proportionally to their payoffs (fitness). Let $x_i(t)$ denote the population share of strategy i and $\pi_i(x)$ its expected payoff against the population state x. The evolution of behavior is given by the replicator dynamic

$$\frac{dx_i}{dt} = x_i\big(\pi_i(x) - \bar{\pi}(x)\big), \qquad \bar{\pi}(x) = \sum_j x_j \pi_j(x). \tag{8.1}$$

In other words, strategies that perform better than average grow in frequency.

In learning models, individual agents repeatedly play the game and update their mixed strategies based on past experience. Under reinforcement learning, the propensity q_i to play action i increases with realized payoff:

$$q_i(t+1) = q_i(t) + \alpha\, \pi_i(t), \qquad x_i(t) = \frac{q_i(t)}{\sum_j q_j(t)}. \tag{8.2}$$

It can be proved [2] that, in large populations and continuous time, this rule converges to the replicator dynamic (8.2). Hence the same mathematical dynamic admits two interpretations:

natural selection in populations and reinforcement learning in minds. Evolution corresponds to adaptation across generations, while learning corresponds to adaptation across experience. Equilibrium concepts (e.g. Nash equilibrium) therefore emerge as long-run stable states of both evolutionary and learning processes. This connection has been discussed in the literature, see for example [2, 7] and, especially, [16].

9 Conclusion

We have studied the Iterated Symmetric Centipede game, which consists of repeated plays of the One-shot Symmetric Centipede. The fact that the OSC is played repeatedly allows the introduction of history-dependent strategies, in particular the Grim trigger strategy, which rewards cooperation and punishes defection. Using the evolutionary game theory approach, and studying both the large-population (using replicator dynamics) and small-population (using Markovian dynamics) cases, we have found that the presence of the defection-punishing Grim strategy ensures the emergence of cooperation.

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A Additional Strategies

We can construct additional evolutionary games in which the players select from a different strategy pool. Here are some examples of additional strategies.

1. Always-m: always stop the game at the m-th stage.
2. Tit-for-Tat: Start by stopping at m-th stage, in all subsequent games stop at the stage that your opponent stoppe din the previous game.
3. ...

B Two Population Replicator Dynamics

The replicator dynamics, as presented in Section 5, assume the existenbce of a single population of players. It follows (as can be seen rather easily) that the one-population RD can only result in *symmetric* NE. As we have seen in Section 4, the ISC also possesses non-symmetric NE. These can be recovered if we assume that *two* populations of players play against each other. I.e., we always slect one player from Population 1 and the other player from Population 2. In this case, letting $x(t)$ (resp. $y(t)$) be the strategy frequency vector of Population 1 (resp. Population 2) at time t, we can postulate that the frequencies evolve according to the *Two-population Replicator Dynamics*:

$$\forall m : \frac{dx_m}{dt} = x_m \left((Cy)_m - x^T C y \right), \tag{B.1}$$

$$\forall m : \frac{dy_m}{dt} = y_m \left(\left(x^T C\right)_m - x^T C y \right). \tag{B.2}$$

Now we can repeat the stability analysis of Section **??** for the system (B.1)-(B.2).